

TWIN-64

Architecture Document

Helmut Fieres
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Introduction

Designers of the seventies and early eighties CPUs almost all used a micro-coded approach with hundreds of instructions. RISC was just starting to enter the stage. Registers were typically not really generic but often had a special function or meaning and many instructions were rather complicated and the designers felt they were useful. The compiler writers often used a small subset ignoring these complex instructions because they were so specialized. The later eighties and nineties shifted to RISC based designs with large sets of registers, fixed word instruction length, a large virtual memory address range and instructions that are in general much more pipeline friendly. Today, processors are highly complex with super-scalar deep pipelines and a multilevel cache hierarchy, just to name a few of the hallmarks of modern CPU design. CPU designs have become complex and large, a single person cannot completely understand. What if there would be a simple design that one person can completely understand ?

Welcome to Twin-64. Twin-64 is a simple 64-bit CPU with a register-memory model and segmented virtual memory. The design is heavily influenced by Hewlett Packard's PA-RISC architecture, which was initially a 32 bit RISC-style load-store machine. Almost all of the key architecture features come from there. However, other processors, such as the Data General MV8000, the DEC Alpha, and the IBM PowerPc, were also influential or at least investigated for their architectural design choices. In addition the Blitz-64 architecture was investigated. Blitz-64, a design developed at Portland State University, offered an interesting approach in that it only support 64-bit signed arithmetic, avoiding common errors from mixing signed and unsigned arithmetic.

While it is not a design goal to build an industry strength, competitive CPU, the CPU should nevertheless be useful and largely follow established common design practices. For example, instructions should not be invented because they seem to be useful, but rather designed with the assembler and compilers in mind. Instruction set design is also CPU design. Register memory architectures were typically micro-coded complex instruction machines. In contrast, Twin-64 instructions will be hard coded and are in general "pipeline friendly", avoiding data and control hazards and stalls where possible.

Where does the name Twin-64 originate from? Obviously the "64" indicates that the CPU is a 64-bit CPU. The "Twin" part will become clearer when implementations of the CPU are described in later chapters. One implementation represents a dual-issue instruction design with two ALUs for computation. Nevertheless, a very basic implementation could just offer a single instruction issues, single ALU design as well.

This document describes the overall architecture, instruction set and runtime model for Twin-64. It is structured into several parts. The first part will give an overview on the system organization and architecture. The major part of the document then presents the instruction set. Memory reference instructions, immediate instructions, branch instructions, computational instructions and system control instructions are described in detail. These chapters are the authoritative reference of the instruction set. The next part will present the major components such as the processor itself, the I/O module architecture and finally the runtime model and calling conventions. An appendix will provide additional details on design and complementary notes.

System Organization

This section will introduce Twin-64 by highlighting the major components and design decision taken.

A Register Memory Architecture

Modern RISC CPUs are typically load/store CPUs and feature a large number of general registers. Operations takes place between registers. Twin-64 follows a slightly different route. There is a smaller number of general registers and in addition to computation between registers, a register-memory model is also supported. There are however no instructions that read and write to memory in one instruction cycle as this would complicate the design considerably.

Twin-64 implements a **register memory architecture** offering few addressing modes for the operand combined with a fixed instruction word length. For most instructions one operand is the register, the other is a flexible operand which could be an immediate, a register content or a memory address from where the data is obtained or placed. The result is placed in the register which is also the first operand. These type of machines are also called two address machines.

In contrast to similar historical register-memory designs, there is no operation which does a read and write operation to memory in the same instruction. For example, there is no "increment memory" instruction, since this would require two memory operations and is in general not pipeline friendly. Memory data access is performed on a machine word, half-word or byte basis. Besides the implicit operand fetch in a computational instruction, there are dedicated memory load / store instructions. In addition to the register-immediate operand and register memory-operand mode, one operand mode supports the three operand model ($R_s = R_a \text{ OP } R_b$), specifying two registers and a result register, which allows to perform three address register for computational operations as well. The machine can therefore operate in a memory register model as well as a load / store register model.

Processor

The Twin-64 processor is only one element of a complete system. A system also includes memory and I/O modules. adapters, and interconnecting busses. The Central Processing Unit (CPU) includes a general register set and machine state registers. To support virtual memory addressing, a hardware translation lookaside buffer (TLB) is included on processors to provide virtual to physical address translations. Instruction and data caches are optional, but for performance reasons a key component of the processor as well. A processor will always emit virtual addresses that need to be translated into physical addresses by the TLB. A Twin-64 system can consists of multiple processors.

Virtual Memory

Twin-64 offers a large global address range, organized in segments. Segment number and offset build the global virtual address. Segments are also associated with access rights and protection identifies. The CPU itself can run in user and privilege mode. Twin-64 emits at the CPU level

virtual addresses. The global virtual memory range is a 52-bit space organized into a 20-bit segment Id and a 32-bit byte offset. There are 2^{20} segments with 2^{32} bytes.

Virtual addresses are translated to physical addresses by the TLB when memory is referenced. For memory management reasons, the virtual address range can also be seen as a set of pages. A segment has thus 2^{18} pages of 16Kb size. A virtual page number combining the segment and page number in that segment is a 2^{38} number. Address translations are made at the page level.

The first segments in the virtual address space are special in that they also represent the physical address range. A virtual address in that range bypasses the TLB and accesses physical memory directly.

A TLB could optionally support different pages sizes in addition to the architected 16Kbyte page size. There are 16Byte, 256Kbyte, 4Mbyte and 64Mbyte pages. It is very common for the operating system and data to be mapped in consecutive physical pages, which are then covered by a larger page size.

Security and Protection Support

Twin-64 implements a protection model along two dimensions. The first dimension is the **access control** and privilege level check. Each page is associated with a page type. There are read-only, read-write, execute and gateway pages. There are two privilege levels, user and supervisor mode. Access type and privilege level form the access control information field, which is checked for each instruction and data memory access.

The second dimension of protection is a **protection Id**, which is the segment Id itself. Twin-64 allows to record a set of segment Id values in the processor control registers. For each access to a segment that has the protection check bit set, one of the protection Id control registers must match the segment Id. If not, a protection violation trap is raised.

Protection Id checking is typically used to form a grouping of access. A good example is the stack data segment, which is accessible at user privilege level. Since the CPU features a global virtual memory address space, every task could access a data stack of another task. With protection Id checking enabled and the segment Id loaded in one of the control registers, only the corresponding user task is allowed to access the stack data segment with a matching protection Id.

Privilege Changes

Instructions can only access data or execute instructions when the privilege level matches the privilege level required. Two or four levels are the most common implementation. Changing between levels is often done with a kind of trap operation to a higher privilege level software, for example the operating system kernel. However, traps are expensive, as they cause a CPU pipeline to be flush when going through a trap handler.

Twin-64 implements gateways for privilege promotion. A special option on the unconditional branch instruction will raise the privilege level when the instruction is executed on a gateway page to the privilege level of that gateway page. Return from a higher privilege level to a code page with lower privilege level, will automatically set the lower privilege level. A branch to a higher privilege level page that is not a gateway page will result in a privilege violation trap.

Debugging Support

A fundamental requirement is the ability to set instruction and data breakpoints. An instruction breakpoint is encountering the **TRAP** instruction. The break instruction will cause an instruction break trap and pass control to the trap handler. Since break instructions are normally not in the instruction stream, they need to be set dynamically in place of the instruction normally found at this instruction address. Upon continuation with a still valid breakpoint for that address, the break point needs to be set again after the execution of the original instruction. To facilitate this mechanism, a bit in the status word will provide an easy way to trap again after the instruction was executed.

Data breakpoints are traps that are taken when a data reference to the page is done. They do not modify the instruction stream. Depending on the data access, the trap could happen before or after the instruction that accesses the data. A write operation is trapped before the data is written, a read instruction is trapped after the original instruction took place.

Input / Output

Twin-64 features a memory-mapped I/O architecture. A portion of the physical address space is dedicated to the I/O subsystem. Within this address range, I/O modules react to memory load and store instructions to their assigned address range. When implementing a driver, the standard page-level protection mechanism is applied during address translation. A driver, typically implemented running in privileged mode, can also grant a user program direct control by mapping an user level I/O page into virtual space without compromising system integrity.

Interrupt System

Interrupts and exceptions are events that occur asynchronously to the instruction execution. Depending on the type, they occur during the execution of an instruction or are checked in between instructions. Example for the former is the arithmetic overflow trap, example for the latter is an external interrupt. Depending on the interrupt or exception type, the instruction is restarted or execution continues with the next instruction.

Twin-64 implements a simple interrupt system. Each processor features an interrupt input register and an interrupt mask register. When an I/O module receives a command, it also receives information which interrupt bit to set and which target module Id to send the interrupt to upon completion of the request. The interrupt data is compared to the interrupt mask register. If the particular bit is enabled and interrupts are in general enabled, the target processor will enter the interrupt handler.

Since causing an interrupt is actually just storing the interrupt request into the processor's interrupt register, I/O modules as well as processors themselves can raise an interrupt at the target processor. Processors can also send interrupts to their peers, implementing low level processor to processor communication.

Instruction Set

Twin-64 features a simple and efficient instruction set. All instructions are of fixed length, which is a 32-bit word for Twin-64. There will be few instructions in total, however, depending on the instruction several options for the instruction execution are encoded to make an instruction more versatile. Great emphasis is placed in that such options do not overly complicated the pipeline and only increase the overall data path length slightly, if at all. Instead of a multitude of addressing modes, typically found in the CISC style CPUs, Twin-64 offers two virtual memory addressing modes with a simple base address - offset calculation model. No indirection or any addressing mode that would require to read a memory item for address calculation is part of the architecture.

The instruction set is divided into five general groups of instructions. The **memory reference** group contains the instructions to load and store to virtual and physical memory. The **immediate** group contains the instructions for building an immediate value up to 64 bit. The **branch** group present the conditional and unconditional instructions. The **computational** instructions perform arithmetic, boolean and bit functions such as bit extract and deposit. Finally, the **control** instruction group contains all instructions for managing hardware elements such as caches and TLBs, register movements, as well as traps and interrupt handling.

Computational instructions

The arithmetic, logical and bit field operations instruction represent the computation type instructions. Twin-64 offers 64-bit signed arithmetic operation only. A numeric overflow always results in a trap.

The binary instructions allow to negate one operand as well as the output. Negating one operand is an efficient way to build complementary bit masks. The bit field instructions will perform bit extraction and deposit as well as double register shifting. These instructions are furthermore the base for all shift and rotate operations.

Immediate instructions

Several instructions have within the instruction word bit fields for representing immediate values. Nevertheless, fixed length instruction word size architectures have one issue in that there is not enough room to embed a full word immediate value in the instruction. Typically, a combination of two or three instructions that concatenate the value from parts is used.

There are furthermore immediate instructions for performing 32-bit address arithmetic. Calculating address will wrap around in the 32-bit address offset and never overflow into the segment Id part.

Memory reference instructions

Memory reference instruction operate on memory and a general register with a unit of transfer being a word, a half-word or a byte. In that sense the architecture is a typical load/store architecture. However, in contrast to a load/store architecture Twin-64 load / store instructions

are not the only instructions that access memory. Being a register/memory architecture, all instructions with an operand field encoding may access memory as well. There are however no instructions that will access data memory reading and writing back an operand in the same instruction.

With the exception of the load reserved and store conditional instruction, memory reference instructions support a base register modification. When the option is set, the base register for an address computation will be adjusted before or after the data access. A negative offset will be added before the address computation, a positive offset after the address computation. Finally, the indexed addressing mode scales the index according to the data type, providing a wider address range reach.

Branch instructions

The control flow instructions are divided into unconditional and conditional branch type instructions. The unconditional branch type instructions allow in addition to altering the control flow the saving of the return point in a general register. Upon subroutine return, there is a branch instruction using this value to return via the general register content.

Often there are code sequence that compare two values and then perform a branch. Conditional instructions allow for comparing two operands and perform a branch if the condition is met.

Branch instructions are further grouped into instruction address relative and global virtual address relative. The former computes an offset using the current instruction address, the latter uses a global virtual address as the base for branch target computation.

System control instructions

The system control instructions perform the control of the CPU resources such as TLB, Cache and so on. There are instructions to access control registers. The TLB modules are controlled by instructions to insert and remove data in these two entities. The cache modules are controlled by flushing and purging data from the cache. There are instructions for generating trap and a general purpose diagnostic instruction for controlling and examining the processor resources. Most of the instructions found in this group require that the processor runs in privileged mode.

Processing Resources

General Registers

Twin-64 features a set of 16 general purposes registers. They are used for data and address values.

	63	0
GR0	Permanent zero	
GR1	Target for ADDIL or General use	
GR2	General use	
	...	
GR14	General use	
GR15	General use	

Figure 1: General registers

Some general registers have a dedicated use. Register zero will always return a zero value when read and a write operation will have no effect. Although there are only 16 general registers in the CPU, implementing such a register greatly simplifies the instruction design, in that it for example provides a target when the result is not needed. General register one is a scratch register and also serves as an implicit target for some instructions. Other general registers may have a dedicated purpose defined in the runtime architecture conventions.

Control Registers

Twin-64 has 16 defined control registers, numbered **CR0** to **CR15**. There are two instructions to move the contents of a control register to a general register and vice versa. A move to a control register will always move the entire 64-bit word. A move to a general register from a control register will move the size of the control register right aligned to the general register. The following table depicts the control registers defined for Twin-64.

	63			32	31			0
CR0	Processor configuration							
CR1	Recovery Counter							
CR2	Reserved							
CR3	Reserved							
CR4	Protection Id 1			W	Protection Id 2			W
CR5	Protection Id 3			W	Protection Id 4			W
CR6	Protection Id 5			W	Protection Id 6			W
CR7	Protection Id 7			W	Protection Id 8			W
CR8	0			IVA			0	
CR9	External Interrupt Enable Mask							
CR10	Interrupt PSW							
CR11	Interrupt Instruction							
CR12	Interrupt Address							
CR13	Scratch Reg 0							
CR14	Scratch Reg 1							
CR15	Scratch Reg 2							

Figure 2: Control registers

Processor Configuration Register

fields with info. one is the actual physical memory size encoded.

one field: 4Bit: Memory Size in 4 Gbyte increments.

```
// ??? work on picture ....
```

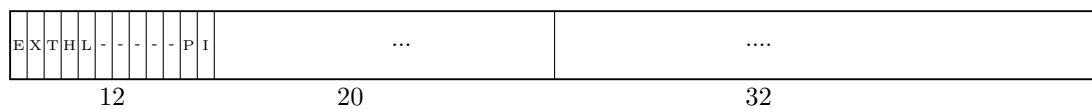


Figure 3: Processor Configuration Register

Table 1: Processor Configuration Register Fields

Field	Purpose
Memory Size	...
Architecture Id	...
Processor Id Id	...
...	...

Recovery Counter

decrements when the PSW status bit is set.

Protection Id Registers

Twin-64 allows to record a set of up to 8 protection Id values in the processor control registers CR4 to CR7 which are accessible to the currently executing process. The protection Id is identical to a segment Id. For each access to a segment that has the protection check bit set, one of the protection Id control registers must match the segment Id. If not, a protection violation trap is raised. The rightmost bit of each of the eight PIDs is the write disable (W) bit. When the bit is set, that PID cannot be used to grant write access. See also the chapter on addressing and access control.

Interrupt Vector Address

The Interruption Vector Address (IVA) control register CR8 contains the address of the base of an array of service procedures assigned to the interruption classes. The lower 14 bits of the IVA are reserved and set to zero. The address written to it must therefore be aligned on a page size. The IVA is always located in the first segment, segment 0. The array of interruption service procedures is indexed by the interruption numbers described in the chapter on interrupts.

External Interrupt Enable Mask

The External Interrupt Enable Mask (EIEM) is stored in CR15). It is 64-bit register containing a bit for each of the 64 external interrupts. A bit cleared in the EIEM masks external interrupts corresponding to that bit position.

Interruption PSW

Upon an interrupt the interrupted instruction address and the processor status bit are saved to this control register.

Interrupt PSW

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Interrupt Argument Registers

The Interrupt Argument Registers (CR11 and CR12) are used to pass an instruction and a virtual address to an interrupt handler. The values in these two control registers for each interruption class are described in the chapter on interrupts.

Scratch Registers

The control registers CR13, CR14, CR15 are scratch registers used by the interrupt handlers or alike. CR12 and CR13 are privileged register only accessible from code running at priv level, CR15 is accessible from any priv level.

Program State Register

The processor state register contains the virtual address of the current instruction and the processor status flags.

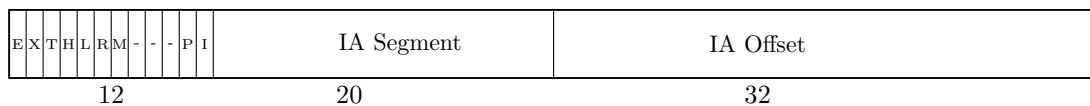


Figure 4: Program State Register

Instruction Address

Processor Status

M I E and so on ... group by user and priv level.

Table 2: Status Field Bits

Bit	Purpose
M	High priority machine check. When set, high-priority machine checks are masked. This bit is set after an HPMC and cleared after all other interruptions.
<i>Continued on next page</i>	

Bit	Purpose
E	Little endian access enable. When set, memory access is in little endian format, else big endian format. Endian support is an optional feature. If not implemented, all access is in big endian format.
R	Recovery counter enable. When set...
X	When set, the processor runs in privileged mode.
H	When set, a higher-privilege transfer trap occurs whenever the following instruction is of a higher privilege.
L	When set, a lower-privilege transfer trap occurs whenever the following instruction is of a lower privilege.
I	When set, external interrupts are enabled.
V	Reserved.
P	Protection identifier validation enable. When set instruction and data references are checked for valid protection identifiers (PIDs). The PRB instruction checks for valid PIDs without causing a trap.
T	When set, taken branch traps are enabled. Any branch taken will result in a trap.
B	The B-bit is set by any taken branch instruction and cleared otherwise.
C	When set, taken branch traps are enabled.
D	Data memory break disable. The D-bit is cleared after the execution of each instruction, except for the RFI instruction which may set it. When set, data memory break traps are disabled. This bit allows a simple mechanism to trap on a data store and then proceed past the trapping instruction.
...	...

Data Types

The fundamental data type is a 64-bit signed integer with the most significant bit being left and the least significant bit being right. All computation is performed on 64-bit signed integer numbers. Memory access is performed on a double, word, half-word or byte basis. Word, Half and Byte data loaded is sign-extended. Store operations just store the corresponding value in memory as is.

All memory addresses are expressed as bytes addresses. Address computation uses a 32-bit unsigned math, i.e. numbers wrap around in a 32-bit space and never cross segment boundaries.

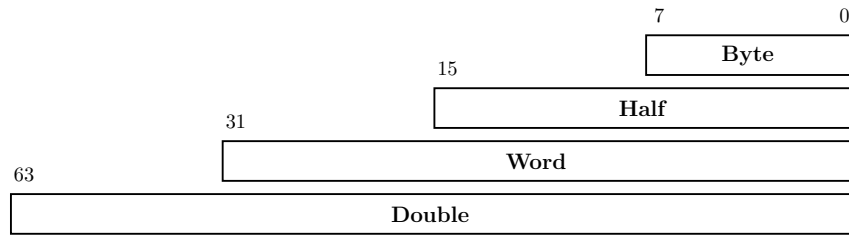


Figure 5: Data types

Byte Ordering

Twin-64 is internally a big-endian machine. It does however support little endian and big endian byte ordering models.

a picture ?

affects only load and stores.

Translation Lookaside Buffer

Virtual address translations are stored in a translation look-aside buffer. The TLB is essentially a copy of the page table entry information that represents the virtual page currently loaded at the physical address. Depending on the hardware implementation, there can be a combined TLB or a separate instruction TLB and data TLB. The virtual page number, VPN, is stored along with attributes and the physical page number, PPN, in the TLB entry. The current architecture allows for a 36-bit physical address space, which is a maximum of 64GB memory space.

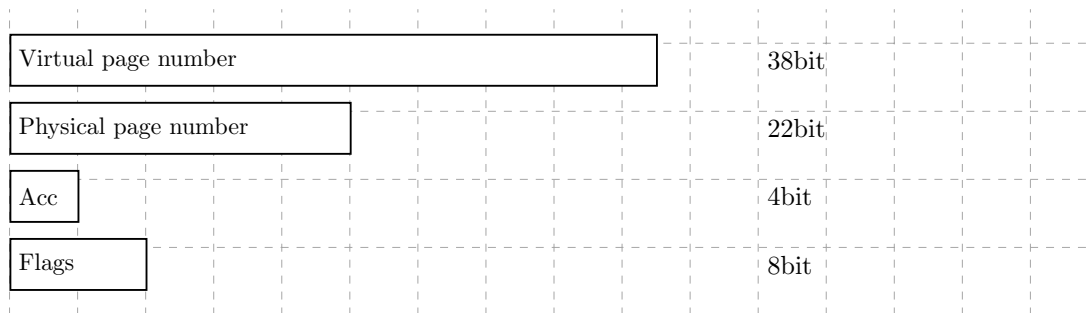


Figure 6: Tlb info

To reduce TLB processing overhead, the TLB is capable of storing virtual address ranges. There is support for a large page size of 16Kb, 256 Kb, 4Mb and 64 Mb.

Table 3: TLB Fields

Field	Purpose
VPN	The virtual page number.
PPN	The physical page number.
Access Rights	The access rights of the page. The field encodes the access type (read, write and execute) and the required priv level for the operation.
V	the entry is valid
L	When set, the TLB entry is locked and will not be selected when replacing an entry.
U	When set, the page represented by the TLB entry is not cached.
P	When set, the page represented by the TLB entry has protection checking enabled.
M	When set, the page represented by the TLB entry has been modified.
B	When set, a branch instruction executed from the page represented by the TLB will cause a branch taken trap.
Page Size	The 2-bit field encodes the supported page size. A value of zero indicates a 16Kbyte page, a value of 1 a 256Kbyte, a value of 2 a 4Mbyte and a value of 3 a 64Mbyte page.

The first segments, segment 0 to 15, are special in that they bypass the TLB entirely. A virtual address in the range 0x00_0000_0000 to 0x03_FFFF_FFFF directly maps to the physical address of the same range. The address range represents physical memory and can only be accessed in privileged mode.

Caches

Twin-64 is a cache visible architecture, featuring cache models for unified or separate instruction and data cache. While hardware directly controls the cache for cache line insertion and replacement, the software has explicit control on flushing and purging cache line entries. Twin-64 caches are virtually indexed, physically tagged caches. To speed up the entire memory access process, the cache uses the virtual address to index into the arrays, which can be done in parallel to the TLB lookup. When the physical address tag in the cache and the physical page address in the TLB match, there will be a cache hit. The Twin-64 cache architecture will not specify the size of cache lines or cache organization. The Twin-64 architecture does not specify a particular cache design and hierarchy model. The architecture just specifies instructions to handle cache operations, such as delete or flushing a cache.

when translation is done, the TLB has the U bit. For the physical segments, the address specifies whether cached or not. I/O range is always uncached. Memory cached.

System Memory

Virtual Address Space

Twin-64 features a 52-bit virtual address range. The address range is organized into a 20-bit segment identifier and a 32-bit offset into that segment. The following figure shows the virtual address and how the segment is selected.

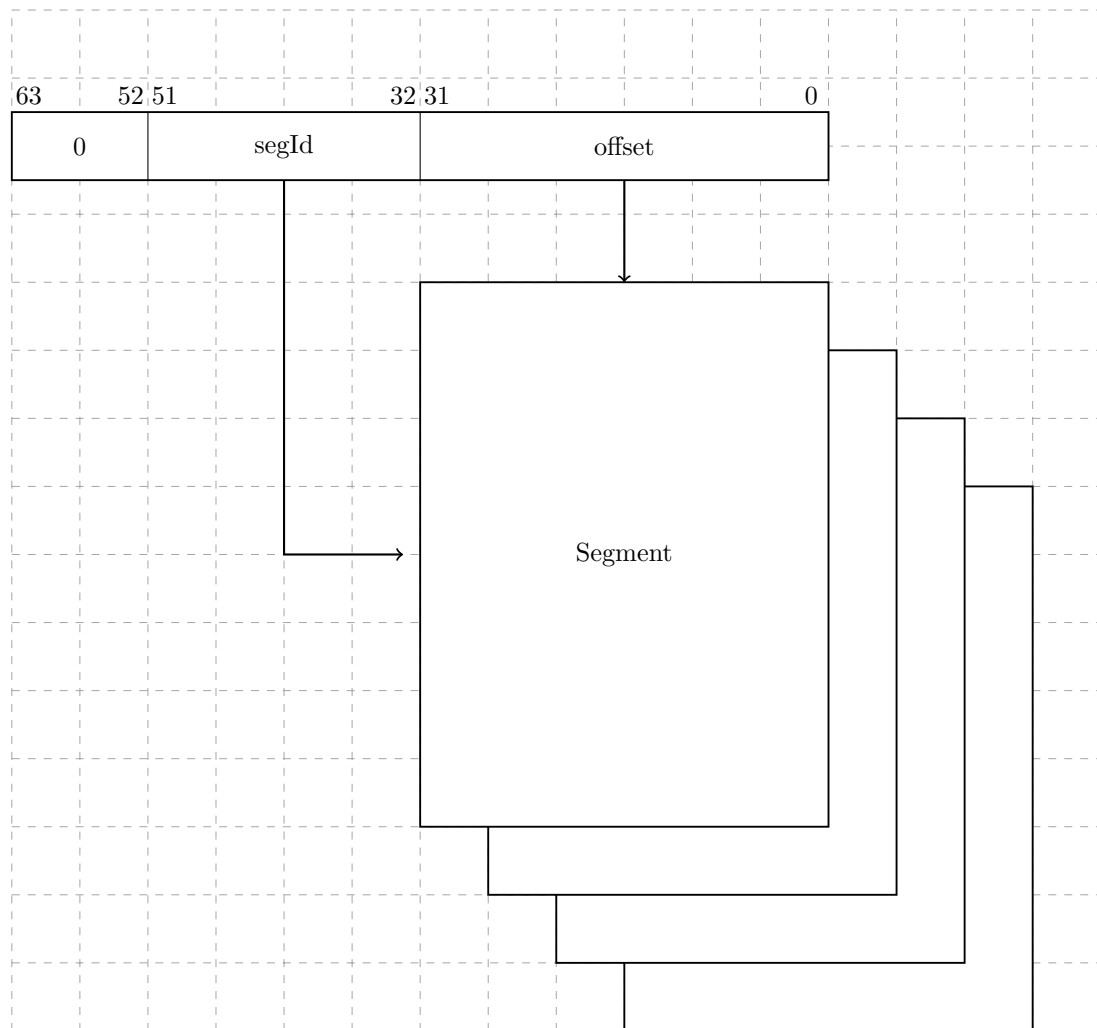


Figure 7: Virtual Memory Address Range

A virtual address to be translated to a physical address. For address translation purposes, the virtual address range can also be seen as a set of virtual pages. The architected page size is 16Kb. A virtual page number is a 38-bit page number with a 14-bit offset into that page.

Physical Address Space

Twin-64 architecture features a 36-bit physical memory address range which allows a maximum physical memory of up to 64 GBytes.

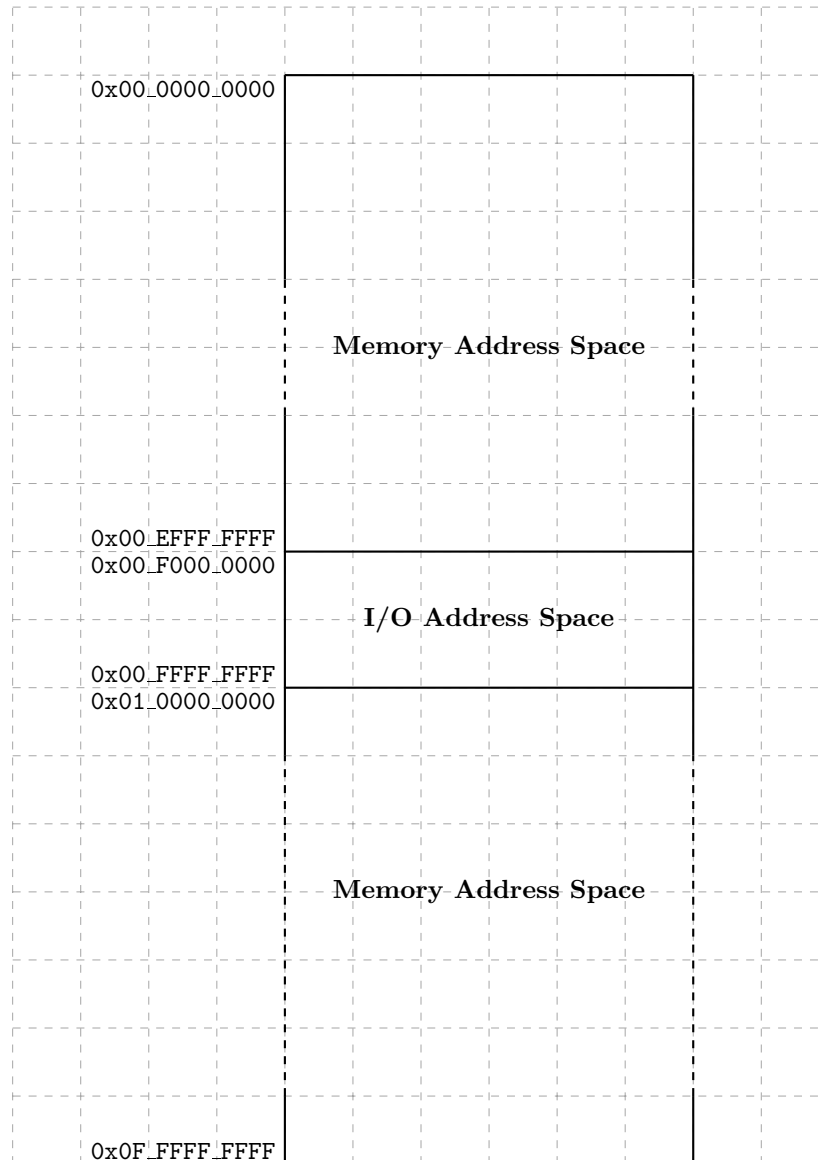


Figure 8: Physical Memory Address Range

Segment Id 0 to 15 directly map to the physical address, without TLB translation and access rights checking. Only privileged mode code can access these segments. Control register zero has a field which encodes the actual memory size installed in a system.

I/O Address Space

Twin-64 features memory mapped I/O architecture. As shown in the overall physical memory layout, the physical memory address range dedicated to I/O modules is located from

0x00_F000_0000 to 0x00_FFFF_FFFF. The following figure depicts the I/O memory address space ranges in more detail.s

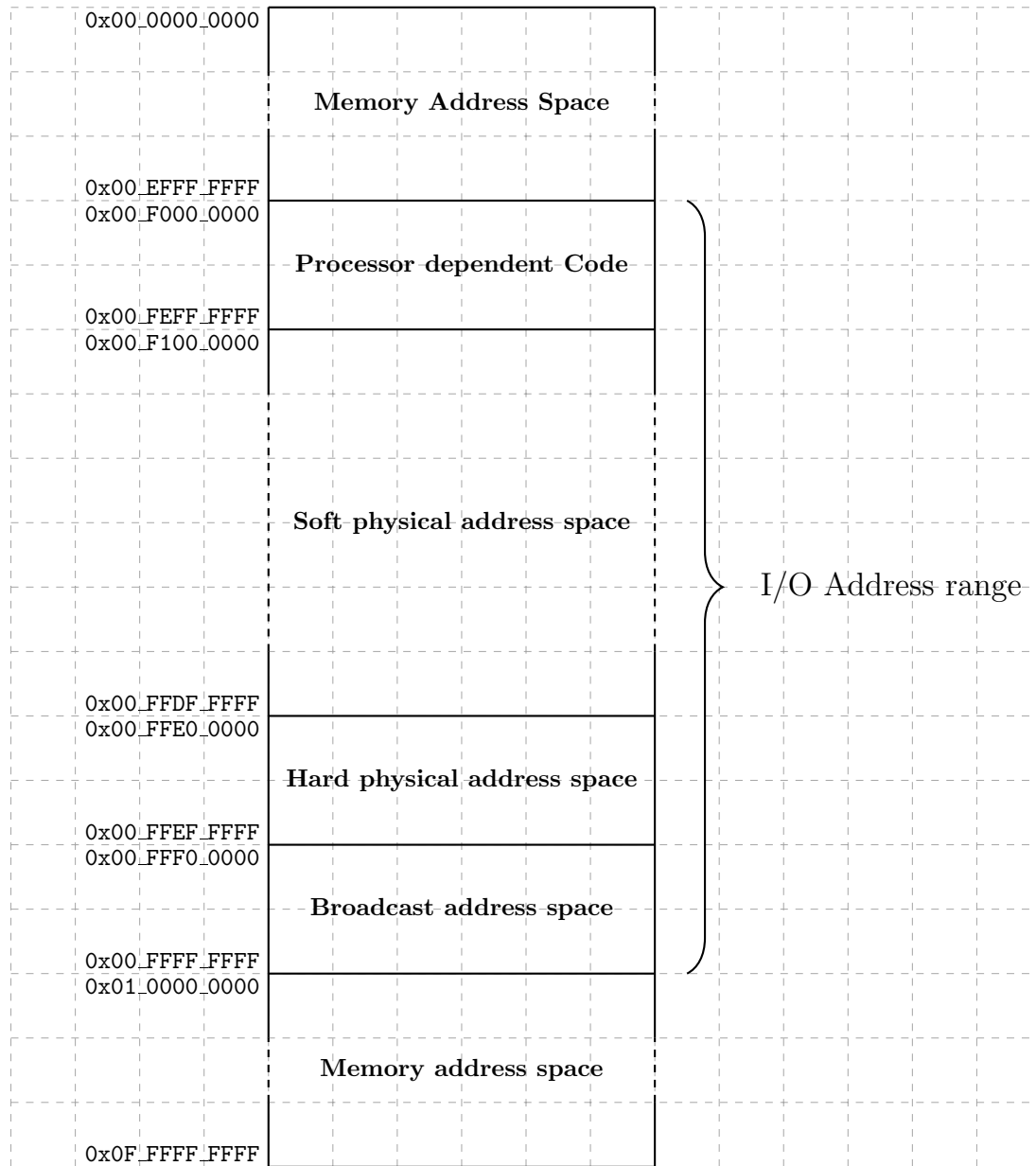


Figure 9: I/O Memory Address Range

The I/O address range is a 256Mbyte area in the first 4 Gbyte segment, i.e. segment 0. The area is divided into several ranges. The first range contains the processor dependent code. There is a set of library functions to manage the processor itself but also provide low-level primitives for the boot process and the operating system.

The **hard physical address** range contains a I/O page for each I/O modules installed. There room for 64 I/O modules.

The **broadcast address** range mirrors the hard physical address range. However, a write to these locations is a write operation to which all I/O modules will react.

The **soft physical addressrange** is an area where space is allocated to an I/O module. The I/O module reacts to a memory access operation in its configured soft physical address range.

Addressing and Access Control

The CPU emits a 52-bit virtual addresses. Each virtual memory access needs to be translated and checked for valid access rights.

Address Translation

Each virtual memory access generated by the processor needs to be translated into a physical address.

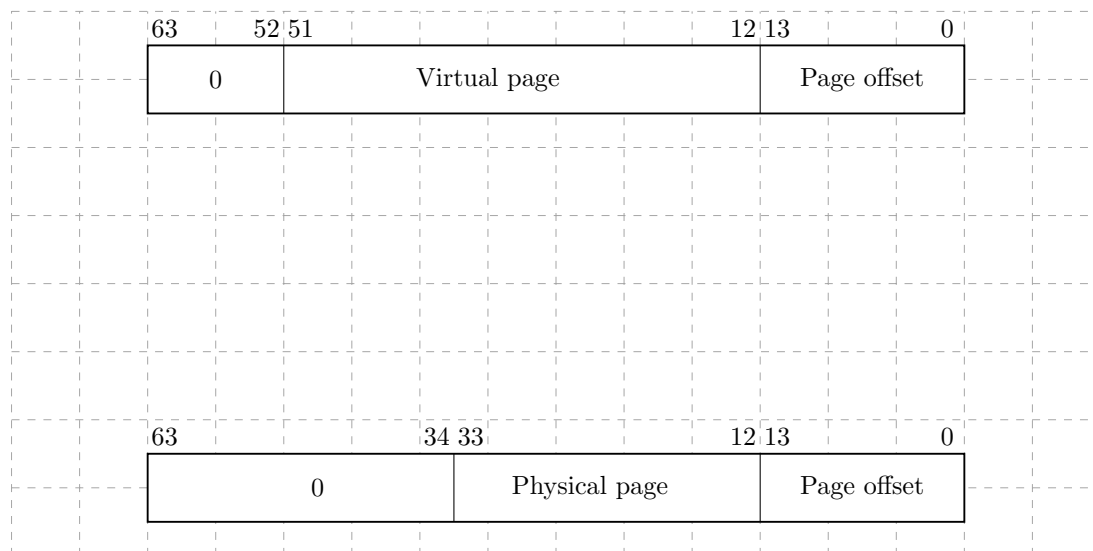


Figure 10: Address translation

virtual address for the first 15 segments directly map to the physical address space.

Access Control

// ??? do not repeat the introducer text already said... go in detail ...

Twin-64 implements a protection model along two dimensions. The first dimension is the **access control** and **privilege level** check. Each page is associated with a page type. There are read-only, read-write, execute and gateway pages. Each memory access is checked to be compatible with the allowed type of access set in the page descriptor. There are two privilege levels, user and supervisor mode. Access type and privilege level form the access control information field, which is checked for each instruction and data memory access. For read access the privilege level us be least as high as the PL1 field, for write access the privilege level must be as least as high as PL2. An exception are the execute and gateway pages, where a write access is only

allowed for privilege code. For execute access the privilege level must be at least as high as PL1 and not higher than PL2. If the instruction privilege level is not within this bounds, a privilege violation trap is raised. If the page type does not match the instruction access type an access violation trap is raised. In both cases the instruction is aborted and a memory protection trap is raised.

picture on access fields

priv levels and operation desired

traps

??? table to show priv levels transitions ?

Table 4: Processor Configuration Register Fields

Field	Purpose
Memory Size	...
...	...

Access Protection

The second dimension of protection is a **protection ID**. Twin-64 allows to record a set of up to 8 protection Id values in the processor control registers **CR4** to **CR7** which are accessible to the currently executing process. The protection Id is identical to a segment Id. For each access to a segment that has the protection check bit set, one of the protection Id control registers must match the segment Id. If not, a protection violation trap is raised.

When protection Id checking is enabled, the eight protection identifiers are compared with the segment Id of the target address to validate an access. The rightmost bit of each of the eight protection Id registers is the write disable (W) bit. When the W bit is set, that protection Id cannot be used to grant write access. When the PSW P-bit is cleared, protection Id and write disable bit checking will not take place.

Protection Id checking is typically used to form a grouping of access. A good example is the stack data segment, which is accessible at user privilege level. Since the CPU features a global virtual memory address space, every task could access a data stack of another task. With protection Id checking enabled and the segment Id loaded into the protection Id assigned control registers, only the corresponding user task is allowed to access the stack data segment with a matching protection Id.

Control Flow

alter instruction flow

no delayed branching concepts

conditional branches

unconditional branches

branching and segments

target address computation

linkage

privilege level changes

traps associated with branches

instruction execution

instruction pipelining

...

Interruptions

Traps

Table 5: Traps

Head 1	Head 2
text 1	text
text 2	text

External Interrupts

picture

mask and request

EIR

EIM

status bit in PSW

Input/Output

Concepts

memory mapped

dedicated address space in physical memory

bus concept

module concept

modules have registers - load / store, however registers do not have to be 64 bit wide.

the system bus features up to 64 modules. that is: 64 HPA pages, need: 1Mb address range.

a module listens to three address ranges

- hard physical address range. a page. privileged.

- soft physical address range. allocated by the module for further space. aligned on a page boundary.

- broadcast address range. Mirrors the system bus range. A write to a register in that range applies to all corresponding registers of a module on the same bus.

modules are processors, memory and I/O cards.

we could keep processor stats in the HPA space

memory module has entire memory as SPA space

I/O elements are 128 bytes, i.e. 16 regs

I/O elements can also be two pages, one priv and one user level, mapped...

Instruction Set Overview

This chapter presents the Twin-64 instruction set. Like typical RISC style instruction sets, it is a fixed length instruction set with very few regular formats. The instructions are group into four categories, arithmetic and logical, load and store, branch and special and system instructions.

Instruction Formats

Twin-64 uses few instruction formats. They are grouped by the number of available register slots and the length of the immediate value field. The **signed immediate S19/special** instruction format provides one register and a 19 bit signed immediate field. This format is typically used by branch instructions.

Grp	OpCode	Reg R	Opt 1	S-IMM-19/special
2	4	4	3	19

The **signed immediate S15/special** instruction format provides two registers and a 15 bit signed immediate field. This format is used by branch instructions as well as instruction which operate on a register and an immediate value.

Grp	OpCode	Reg R	Opt 1	Reg B	S-IMM-15/special
2	4	4	3	4	15

The **signed immediate S13/special** instruction format provides two registers, an additional option field and a 13 bit signed immediate field. Some instruction use the S-IMM-13 field for special encodings instead of a signed value. This format is used by memory reference instructions where the address is formed by a base register and a 13-bit signed offset.

Grp	OpCode	Reg R	Opt 1	Reg B	Opt 2	S-IMM-13/special
2	4	4	3	4	2	13

The **register/signed immediate S9/special** instruction format provides three registers, an additional option field and a 9 bit signed immediate field. Some instruction use the S-IMM-9 field for special encodings instead of a signed value. This instruction format is used for all computational instructions where three registers are needed.

Grp	OpCode	Reg R	Opt 1	Reg B	Opt 2	Reg A	S-IMM-9/special
2	4	4	3	4	2	4	9

The **unsigned immediate U20** format is used by the immediate instruction group to provide a 20bit unsigned value. This value is then placed at certain positions in the register target Reg R.

Grp	OpCode	Reg R	0	U-IMM-20
2	4	4	2	20

Arithmetic Instructions

The arithmetic instruction operate on two signed 64-bit operands and stores the result to the target register. There are two instruction layouts, signed immediate S15 and register based. The **immediate operand instruction format** will add the sign extended 15-bit value to **RegB**. The **register instruction format** type instruction uses two registers as input operands, performs the operation and stores the result in a third register. The following figure shows the instruction layout.

The **ADD** and **SUB** instructions perform signed integer 64-bit arithmetic with a numeric overflow resulting in an overflow trap. There are no operations on unsigned quantities. The **SHLxA** and **SHRxA** perform an arithmetic left shift operation and add an operand to the shifted input, storing the result in the target register. The **CMP** compares two registers for being equal, not equal, less than and less or equal than. The result of this comparison is stored in the target register.

Bit Operations Instructions

The bit operation instruction fall into two groups. The **AND**, **OR** and **XOR** instructions implement the logical operations as their name suggest. Like the arithmetic instructions, there is an immediate instruction and a register instruction layout.

The second group implements the bit manipulation operations. The **EXTR** instruction extracts a bit field from a register and places the field in another register. Optionally, the extracted field is signed extended. The **DEP** instruction deposits a bit field into a target register. The **DSR** instruction concatenates two general registers, shifts the 128-bit quantity to the right and stores the lower 64 bits in a target register.

Immediate Instructions

The **LDI**, and **ADDIL** are instruction for forming large constants. With a 32-bit instruction word, even 32-bit constants are not possible in one instruction. The **LDI** instruction has a qualifier to load a constant value at a defined position in the target register. The **opt** field allow to select fields in the target register where the constant value is stored. With a short instruction sequence of any 64-bit value can be loaded.

The **ADDIL** instruction adds the upper portion of a 32-bit immediate value to the lower 32-bit half of a general register. This instruction is used primarily for address arithmetic to allow reaching any location in a segment. The **LDO** instruction loads an offset into a general register. The address loaded is formed by adding the signed immediate value to the base register. Address arithmetic is an unsigned 32-bit arithmetic with wrapping around in the 32-bit range.

Memory Reference Instructions

Twin-64 is a register memory architecture. It has the common load/store instructions as well as instructions that combined an operation with a memory reference. Memory reference instructions feature two addressing modes. The first is the base register and offset instruction. the second the base register and register index instruction.

The **indexed instruction format** will load the content of memory address formed by adding the scaled immediate 13-bit field to the base register **RegB** and add the data value to **RegA**. The **dw** field indicates the data value width. A **dw** value of zero loads an 8-bit, a value of 1 a 16-bit value, value of 2 a 32-bit value and a value of 3 a 64-bit value. In any case, the data value fetched is signed extended to a 64-bit value. In addition, the **dw** field will scale the offset value by left shifting the immediate field according to the data width. A **dw** value of 3, for example, will load an double value from an address formed by shifting the offset index in the instruction by three bits before adding to the base register.

The **register indexed instruction format** will load the content of memory address formed by adding **RegX** to the base register **RegB**. The **dw** field to specify the data value width as in the indexed instruction format. The offset in **RegX** is interpreted as a byte offset, independent of the actual **dw** field value. In both instruction cases, the data value fetched is signed extended to a 64-bit value.

The **LD** and **ST** instruction are the load and store instructions. In addition to the two addressing modes presented above, they furthermore feature a base register address modification. Depending on the offset sign, the base register is updated before or after the address computation with the effective address computed. See the instruction description for more details.

The **LDR** and **STC** instruction implement the base support for a pipeline friendly method for semaphore operations. They only support the indexed instruction mode on a double word. Also, there is no base register modification option.

The **ADD**, **SUB**, **AND**, **OR**, **XOR**, **CMP** instructions perform the respective operation as described above with one operand being fetched from memory using the indexed addressing modes described above.

Control Flow Instructions

Control flow is implemented through a set of branch instructions. They can be classified into unconditional and conditional instruction branches.

Unconditional branches fetch the next instruction address from the computed branch target. The **B** instruction performs an instruction relative branch, the **BR** instruction a base register relative branch. Both optionally return the return address in a general register.

The **BV** instruction is a vectored branch. The content of a general register is added to a base register, forming branch target.

The instructions **BB**, **CBR**, **MBR** and **ABR** implement the conditional branch instructions. If the condition is met a branch to the target address is performed. The target address is always a local address and the offset is instruction address relative. Conditional branches adopt a static

prediction model. Forward branches are assumed not taken, backward branches are assumed taken.

All branch address arithmetic is a 32-bit unsigned arithmetic with wraparound. Adding branch offsets will not overflow into the segment Id portion of the address..

System Instructions

The final instruction group is the **special/system** instruction group. The system control instructions control the CPU resources such as TLB, Cache and so on. Most of the instructions found in this group require that the processor runs in privileged mode.

The **MFCR** and **MTCR** instructions are used to transfer data to and from a control register. Access to some control registers requires privileged mode. The **MFIA** instruction accesses the current instruction address offset, a key instruction for position independent code aspects. The **RSM** and **SSM** instructions allow for setting or clearing an individual accessible bit of the status part of the processor state.

The **LDPA**, **PRBR** and **PRBW** instructions are used to obtain information about the virtual memory system. The LDPA instruction returns the physical address of the virtual page, if present in memory. The PRB instruction tests whether the desired access mode to an address is allowed.

The TLB and Caches are managed by the **ITLB**, **PTLB**, **FCA** and **PCA** instruction. The ITLB and PTLB instructions insert into the TLB and remove translations from the TLB. The FCA and PCA instructions manages the instruction and data cache, featuring to flush and / or just purge a cache line. There is no instruction that inserts data into a cache as cache insertion is completely controlled by hardware.

The **RFI** instruction is used to restore a processor state from the control registers. Optionally, general registers that have been stored into the reserved control registers will be restored.

The **DIAG** instruction is a control instructions to issue hardware specific implementation commands. Example is the memory barrier command, which ensures that all memory access operations are completed after the execution of this DIAG instruction.

The **TRAP** instruction ...

Instruction Set Descriptions

This chapter presents the Twin-64 instruction set. Like typical RISC style instruction sets, it is a fixed length instruction set with very few regular formats. The instructions are group into four categories, arithmetic and logical, load and store, branch and special and system instructions.

The instructions described in this chapter contains the instruction word layout, a short description of the instruction and also a high level pseudo code of what the instruction does. The pseudo code routines used in the instruction descriptions are listed in the appendix. Each instruction operation is described in a C-style pseudo code operation. The pseudo code uses a register notation describing by their class and index. For example, GR[5] labels general register 5 and CR[1] the control register number 1.

In addition, some register also have dedicated names, such as for example the shift amount control register, labelled "SAR". A subfield of the instruction is selected by adding a "." and the field name in square brackets. For example the interrupt enable bit in the PSW register is described as "PSW.[E]".

Additional options for an instruction will be specified by a "." followed by a sequence of characters in the assembler syntax. Each character is one of the defined options for the instruction. As an example, the AND instruction can negate the result. The assembler notation would be "AND.N ...". The order of the individual characters does not matter. The defined options are listed as a list of characters.

Reserved fields in an instruction word are reserved for future enhancements and should be filled with a zero. The instruction explicitly shows the numeric value of these fields. Any other value at that place will result in an undefined behavior.

ABR - Add and branch

Syntax

ABR.<cond> RegR, RegB, target

where cond is EQ, LT, NE or LE.

Format

The ABR instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description

The ABR instruction ...

Operation

...

Exceptions

None.

Notes

None.

ADD Integer Addition

Syntax

```
ADD RegR, RegB, RegA
ADD RegR, RegB, SignedImmed15

ADD[.D/W/H/B] RegR, SignedImmed13( RegB )
ADD.D/W/H/B] RegR, RegX ( RegB )
```

Format

The ADD instruction uses the register, the immediate-15, the indexed and the register indexed instruction formats.

0	1	Reg R	0	Reg B	0	Reg A	0
2	4	4	3	4	2	4	9

0	1	Reg R	1	Reg B	S-IMM-15
2	4	4	3	4	15

1	1	Reg R	0	Reg B	dw	S-IMM-13
2	4	4	3	4	2	13

1	1	Reg R	1	Reg B	dw	Reg X	0
2	4	4	3	4	2	4	9

Description

The ADD instruction adds two signed 64-bit operands and stores the result to the target register **RegR**. The instruction supports the immediate, the register and the two memory reference instruction formats. An arithmetic overflow results in a numeric overflow trap. The indexed instruction formats may cause a memory reference associated trap.

Operation

```
RegR <- RegB + RegA (register format)
RegR <- RegB + immOperand( ) (immediate format)
RegR <- RegR + memOperand( ) (indexed formats)
```

Exceptions

Integer Overflow Trap
Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

ADDIL Add immediate left

Syntax `ADDIL RegR, val`

Format The ADDIL instruction uses the immediate-20 instruction format.

0	9	Reg R	0	val
2	4	4	2	20

Description The ADDIL instruction ...

Operation `RegR <- RegR + ( val << 12 );`

Exceptions None.

Notes None.

AND Boolean Operation

Syntax

```
AND[.C/N] RegR, RegB, RegA
AND[.C/N] RegR, RegB, SignedImmed15

AND[.C/N/D/W/H/B] RegR, SignedImmed13( RegB )
AND[.C/N/D/W/H/B] RegR, RegX ( RegB )
```

Format

The AND instruction uses the register, the immediate-19, the indexed and the register indexed instruction formats.

0	3	Reg R	N	C	0	Reg B	0	Reg A	0
2	4	4	3			4	2	4	9

0	1	Reg R	N	C	1	Reg B	S-IMM-15
2	4	4	3			4	15

1	1	Reg R	N	C	0	Reg B	dw	S-IMM-13
2	4	4	3			4	2	13

1	1	Reg R	N	C	1	Reg B	dw	Reg X	0
2	4	4	3			4	2	4	9

Description

The AND instruction performs the binary AND operation on two 64-bit operands and stores the result to the target register **RegR**. Upon input the general register **regB** can be negated. The result can also be negated by setting the "C" bit. The result can be negated by setting the "N" bit. The instruction supports the immediate, the register and the two memory reference instruction formats. The boolean operation itself does not cause any traps. The indexed formats may cause a memory reference associated trap.

Operation

```
if ( instr.[C] RegB <- not( RegB );
RegR <- RegB & RegA (register format)
RegR <- RegB & immOperand( ) (immediate format)
RegR <- RegR & memOperand( ) (indexed formats)
if ( instr.[N] RegR <- not( RegR );
```

Exceptions

Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

B branch unconditional

Syntax B[.G] target [, RegR]

Format The B instruction uses the immediate-19 format.

2	1	Reg R	0	G	target
2	4	4	2	3	19

Description The B instruction ...

Operation RegR <- addOfs32(PSW.[IA], 4); PSW.[IA] <- addOfs32(
PSW.[IA], target);

Exceptions Instruction Address Trap Instruction Privilege Violation Trap Instruc-
tion TLB Miss Trap

Notes None.

BB branch on bit

Syntax

BB [.T/F] ofs, pos
BB [.T/F] ofs, SAR

Format

The BB instruction uses the the immediate-9 instruction formats

2	4	Reg R	0	A	V	pos	ofs
2	4	4	1	1	1	6	9

Description

The BB instruction ...

Operation

```
if ( instr.[A] ) tmp <- SAR;  
else ( tmp <- instr.[pos];  
if ( checkBit( RegR, instr.[V] ) PSW.[IA] <- addOfs(  
PSW.[IA], ofs );  
else PSW.[IA] <- addOfs( PSW.[IA], 4 );
```

Exceptions

tbd...

Notes

None.

BR branch register

Syntax BR RegB [, RegR]

Format The BR instruction uses the immediate-15 instruction format.

2	2	Reg R	0	Reg B	0
2	4	4	3	4	15

Description The BR instruction ...

Operation `RegR <- addOfs32( PSW.[IA], 4 );      PSW.[IA] <- addOfs32( RegB, target );`

Exceptions None.

Notes None.

BV branch vectored

Syntax

```
RegR <- addOfs32( PSW.[IA], 4 );  
PSW.[IA] <- addOfs32( RegB, RegX );
```

Format

The BV instruction uses the XXX instruction formats.

2	7	Reg R	0	M	1	Reg B	0	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description

The BV instruction ...

The indexed formats may cause a memory reference type trap.

Operation

...

Exceptions

...

Notes

None.

CBR compare and branch

Syntax

CBR.<cond> RegR, RegB, target

where cond is EQ, LT, NE or LE.

Format

The CBR instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description

The CBR instruction ...

Operation

...

Exceptions

None.

Notes

None.

CMP Integer Compare

Syntax

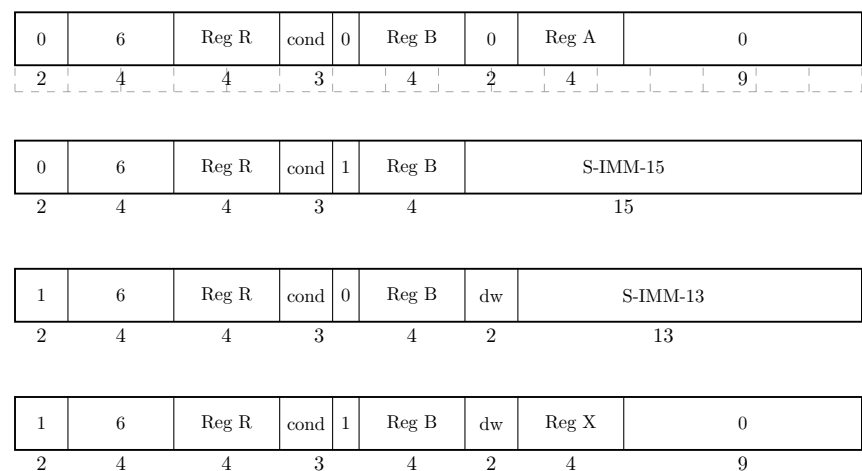
```
CMP.<cond> RegR, RegB, RegA
CMP.<cond> RegR, RegB, SignedImmed15

CMP.<cond>[.C/N/D/W/H/B] RegR, SignedImmed13( RegB )
CMP.<cond>[.C/N/D/W/H/B] RegR, RegX ( RegB )
```

where cond_i is EQ, LT, NE or LE.

Format

The CMP instruction uses the register, the immediate-15, the indexed and the register indexed instruction formats.



Description

The **CMP** instruction performs an integer comparison of the two 64bit signed operands and depending on the comparison stores a 0 or 1 in the target register. The indexed formats may cause a memory reference associated trap.

The condition field encodes the comparison options. The options are a 0 for equal, a 1 for less than, a 2 for not equal and a 3 for less than or equal.

Operation

```
cond <- instr.[cond];
RegR <- cmpOp( cond, RegB, RegA ) (register format)
RegR <- cmpOp( cond, RegB, immOperand( )) (immediate format)
RegR <- cmpOp( cond, RegR, memOperand( )) (indexed formats)
```

Exceptions

Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

DEP Deposit Bit Field

Syntax

```
DEP[.Z/I] RegR, RegB, pos, len  
DEP[.Z/I] RegR, RegB, SAR, len
```

Format

The DEP instruction uses the special-13 instruction format with special fields.

0	7	Reg R	0	Reg B	I	A	Z	pos	len
2	4	4	3	4	3			4	4

Description

The DEP instruction deposits the bit field with the specified length in general register RegB into the general register RegR at the specified position. The position field specifies the rightmost bit for the bit field to deposit. The length field specifies the bit size of the field to deposit. The Z bit clears the target register before storing the bit field. If the A bit is set, the shift amount control register is used for obtaining the position value. The I bit specifies that the RegA field contains an immediate value instead of a register.

Operation

```
if ( instr.[Z] ) RegR <- 0;  
if ( instr.[A] ) pos <- shamr;  
else pos <- instr.[pos];  
if ( instr.[I] )  
  RegR <- depositImm( RegR, RegB, pos, instr.[6:0] );  
else  
  RegR <- depositField( RegR, RegB, pos, instr.[6:0] );
```

Exceptions

None

Notes

None.

DIAG - ...

Syntax

DIAG ...

Format

The DIAG instruction uses the XXX instruction formats.

2	7	Reg R	0	M	1	Reg B	0	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description

The DIAG instruction ...

The indexed formats may cause a memory reference type trap.

Operation

...

Exceptions

...

Notes

None.

DSR Double Shift Right

Syntax

DSR RegR, RegB, RegA, amt
DSR RegR, RegB, RegA, SAR

Format

The DSR instruction uses the special-9 instruction format.

0	7	Reg R	3	Reg B	0	A	Reg A	0	amt
2	4	4	3	4	2	4	2	6	

Description

The DSR instruction concatenates the general registers RegB and RegA and performs a right shift operation. Register RegB is the left part, register RegA the right part. The lower 64 bits of the result are stored in the general register RegR. The amount field contains the shift amount. If the A bit is set, the shift amount is taken from the shift amount control register.

Operation

```
if ( instr.[A] ) tmp <- shamReg;  
else tmp <- instr.[amt];  
RegR <- doubleShiftR( RegB, RegA, tmp );
```

Exceptions

None.

Notes

None.

EXTR Extract Bit Field

Syntax

```
EXTR[.S] RegR, RegB, pos, len  
EXTR[.S] RegR, RegB, SAR, len
```

Format

The EXTR instruction uses the special-9 instruction format with special fields.

0	7	Reg R	0	Reg B	0	A	S	pos	len
2	4	4	3	4	2			4	9

Description

The EXTR instruction performs a bit field extract specified by the position and length instruction field from general register RegB. The position field specifies the rightmost bit of the bitfield to extract. The length field specifies the bit size if the field to extract. The extracted bit field is stored right justified in the general register RegR. If the S bit is set, the result is sign extended. If the A bit is set, the shift amount control register is used for obtaining the position value instead of the length instruction field.

Operation

```
if ( instr.[A] tmp <- shamt;  
else tmp <- instr.[pos];  
RegR <- extractField( RegB, pos, len );  
if ( instr.[S] ) RegR <- signExtend( RegR, pos );
```

Exceptions

None

Notes

None.

FCA - flush from cache

Syntax FCA RegR, RegB

Format The PTLB instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The FCA instruction ...

Operation ...

Exceptions None.

Notes None.

ITLB insert TLB entry

Syntax ITLB ...;

Format The ITLB instruction uses the XXX instruction formats.

2	7	Reg R	0	M	1	Reg B	0	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description The ITLB instruction ...

The indexed formats may cause a memory reference type trap.

Operation ...

Exceptions ...

Notes None.

LD Load register

Syntax

LD[.D/W/H/B] RegR, SignedImmed13(RegB)
LD.D/W/H/B] RegR, RegX (RegB)

Format

The LD instruction uses the indexed and the register indexed instruction formats.

1	7	Reg R	0	M	0	Reg B	dw	S-IMM-13
2	4	4	1	1	1	4	2	13

1	7	Reg R	0	M	1	Reg B	dw	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description

The LD instruction ...

The indexed formats may cause a memory reference type trap.

Operation

RegR <- memLoad(memOperand()) (indexed formats)

Exceptions

Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

LDI Load immediate

Syntax LDI [.L/S/U] RegR, val

Format The LDI instruction uses the immediate-20 instruction format.

0	9	Reg R	opt	val
2	4	4	2	20

Description The LDI instruction ...

Operation

$$\text{RegR} \leftarrow ((\text{val} \& 0\text{x}\text{FFFFF}) \ll 12); \text{ (.L option)}$$
$$\text{RegR} \leftarrow ((\text{val} \& 0\text{x}\text{FFFFF}) \ll 32); \text{ (.S option)}$$
$$\text{RegR} \leftarrow ((\text{val} \& 0\text{x}\text{FFF}) \ll 52); \text{ (.U option)}$$

Exceptions None.

Notes None.

LDO load offset

Syntax LDO RegR, ofs (RegB)

Format The LDO instruction uses the immediate-15 instruction format.

0	10	Reg R	0	Reg B	ofs
2	4	4	3	4	15

Description The LDO instruction ...

Operation $\text{RegR} \leftarrow \text{RegR} + \text{ofs};$

Exceptions None.

Notes None.

LDR Load register reserved

Syntax LDR RegR, SignedImmed13(RegB)

Format The LDR instruction uses the indexed instruction format.

1	9	Reg R	0	Reg B	3	S-IMM-13
2	4	4	0	4	2	13

Description The LDR instruction ...

The indexed formats may cause a memory reference type trap.

Operation ??? fix ...

RegR <- memLoad(memOperand()) (indexed formats)

Exceptions Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes None.

LPA - load physical address

Syntax LPA ...

Format The BV instruction uses the XXX instruction formats.

2	7	Reg R	0	M	1	Reg B	0	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description The LPA instruction ...

The indexed formats may cause a memory reference type trap.

Operation ...

Exceptions ...

Notes None.

MBR move and branch

Syntax

MBR.<cond> RegR, RegB, target

where jcond_i is EQ, LT, NE or LE.

Format

The MBR instruction uses the xxx instruction format.

0	6	Reg R	cond	0	Reg B	ofs
2	4	4	2	1	4	15

Description

The MBR instruction ...

Operation

...

Exceptions

None.

Notes

None.

MFIA - Move from instruction address register

Syntax MFIA RegR

Format The MFIA instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The MTCR instruction ...

Operation ...

Exceptions None.

Notes None.

MFCR Move from Control register

Syntax MFCR RegR, CRegB

Format The MFCR instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The CBR instruction ...

Operation ...

Exceptions None.

Notes None.

MTCR Move to Control register

Syntax MTCR RegR, CRegB

Format The MFCR instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The MTCR instruction ...

Operation ...

Exceptions None.

Notes None.

NOP - No operation

Syntax NOP]

Format The RFI instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The NOP instruction ...

Operation ...

Exceptions None.

Notes None.

OR Boolean Operation

Syntax

```
OR[.C/N] RegR, RegB, RegA
OR[.C/N] RegR, RegB, SignedImmed15

OR[.C/N/D/W/H/B] RegR, SignedImmed13( RegB )
OR[.C/N/D/W/H/B] RegR, RegX ( RegB )
```

Format

The OR instruction uses the register, the immediate-19, the indexed and the register indexed instruction formats.

0	4	Reg R	N	C	0	Reg B	0	Reg A	0
2	4	4	3		4	2	4		9

0	4	Reg R	N	C	1	Reg B	S-IMM-15
2	4	4	3		4		15

1	4	Reg R	N	C	0	Reg B	dw	S-IMM-13
2	4	4	3		4	2		13

1	4	Reg R	N	C	1	Reg B	dw	Reg X	0
2	4	4	3		4	2	4		9

Description

The OR instruction performs the binary OR operation on two 64-bit operands and stores the result to the target register **RegR**. Upon input the general register **regB** can be negated. The result can also be negated by setting the "C" bit. The result can be negated by setting the "N" bit. The instruction supports the immediate, the register and the two memory reference instruction formats. The boolean operation itself does not cause any traps. The indexed formats may cause a memory reference associated trap.

Operation

```
if ( instr.[C] RegB <- not( RegB );
RegR <- RegB | RegA (register format)
RegR <- RegB | immOperand( ) (immediate format)
RegR <- RegB | memOperand( ) (indexed formats)
if ( instr.[N] RegR <- not( RegR );
```

Exceptions

Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

PCA - purge from cache

Syntax PCA RegR, RegB

Format The PTLB instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The PCA instruction ...

Operation ...

Exceptions None.

Notes None.

PRB - probe virtual address

Syntax PRB ...

Format The PRB instruction uses the XXX instruction formats.

2	7	Reg R	0	M	1	Reg B	0	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description The PRB instruction ...

The indexed formats may cause a memory reference type trap.

Operation ...

Exceptions ...

Notes None.

PTLB - purge from TLB

Syntax PTLB RegR, RegB

Format The PTLB instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The PTLB instruction ...

Operation ...

Exceptions None.

Notes None.

RFI - Return from interrupt

Syntax RFI [RegR]

Format The RFI instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The RFI instruction ...

Operation ...

Exceptions None.

Notes None.

RSM - Reset system mask

Syntax RSM RegR,

Format The MFIA instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The RSM instruction ...

Operation ...

Exceptions None.

Notes None.

SHLxA Shift left and add

Syntax

SHLxA RegR, RegB, RegA
SHLxA RegR, RegB, SignedImmed13

Format

The SHLxA instruction uses the register and the immediate instruction formats.

0	8	Reg R	amt	0	Reg B	0	Reg A	0
2	4	4	1 2		4	2	4	9

1	8	Reg R	amt	0	Reg B	2	S-IMM-13
2	4	4	1 2		4	2	13

Description

The SHLxA instruction ...

Operation

RegR $\leftarrow$ (RegB $\ll$ amt) + RegA (register format)
RegR $\leftarrow$ (RegB $\ll$ amt) + immOperand() (immediate format)

Exceptions

Integer Overflow Trap

Notes

None.

SHRxA Shift right and add

Syntax

SHRxA RegR, RegB, RegA
SHRxA RegR, RegB, SignedImmed13

Format

The SHRxA instruction uses the register and immediate instruction formats.

0	8	Reg R	amt	0	Reg B	0	Reg A	0
2	4	4	1 2		4	2	4	9

1	8	Reg R	amt	0	Reg B	2	S-IMM-13
2	4	4	1 2		4	2	13

Description

The SHRxA instruction ...

Operation

RegR <-(RegB >> amt) + RegA (register format)
RegR <- (RegB >> amt) + immOperand() (immediate format)

Exceptions

Integer Overflow Trap

Notes

None.

SSM - Set system mask

Syntax SSM RegR,

Format The MFIA instruction uses the xxx instruction format.

2	5	Reg R	cond	0	Reg B	ofs
2	4	4	3		4	15

Description The SSM instruction ...

Operation ...

Exceptions None.

Notes None.

ST Store register

Syntax

ST[.D/W/H/B] RegR, SignedImmed13(RegB)
ST.D/W/H/B] RegR, RegX (RegB)

Format

The ST instruction uses the indexed and the register indexed instruction formats.

1	8	Reg R	0	M	0	Reg B	dw	S-IMM-13
2	4	4	1	1	1	4	2	13

1	8	Reg R	0	M	1	Reg B	dw	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description

The ST instruction ...

The indexed formats may cause a memory reference type trap.

Operation

RegR <- memLoad(memOperand()) (indexed formats)

Exceptions

Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

STC Store register conditional

Syntax STC RegR, SignedImmed13(RegB)

Format The STC instruction uses the indexed instruction format.

1	8	Reg R	0	Reg B	3	S-IMM-13
2	4	4	0	4	2	13

Description The ST instruction ...

The indexed formats may cause a memory reference type trap.

Operation $\text{RegR} \leftarrow \text{memLoad}(\text{memOperand}())$ (indexed formats)

Exceptions

- Data Alignment Trap
- Memory Access Violation Trap
- Memory Protection Violation Trap
- Data TLB Miss Trap

Notes None.

SUB Integer Subtraction

Syntax

```
SUB RegR, RegB, RegA
SUB RegR, RegB, SignedImmed15

SUB[.D/W/H/B] RegR, SignedImmed13( RegB )
SUB.D/W/H/B] RegR, RegX ( RegB )
```

Format

The SUB instruction uses the register, the immediate, the indexed and the register indexed instruction formats.

0	2	Reg R	0	Reg B	0	Reg A	0
2	4	4	3	4	2	4	9

0	2	Reg R	1	Reg B	S-IMM-15		
2	4	4	3	4	15		

1	2	Reg R	0	Reg B	dw	S-IMM-13	
2	4	4	3	4	2	13	

1	2	Reg R	1	Reg B	dw	Reg X	0
2	4	4	3	4	2	4	9

Description

The SUB instruction subtracts a signed 64-bit operand from a general register **RegB** and stores the result to the target register **RegR**. The instruction supports the immediate, the register and the two memory reference instruction formats. An arithmetic overflow results in a numeric overflow trap. The indexed formats may cause a memory reference associated trap.

Operation

```
RegR <- RegB - RegA (register format)
RegR <- RegB - immOperand( ) (immediate format)
RegR <- RegR - memOperand( ) (indexed formats)
```

Exceptions

Integer Overflow Trap
Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

TRAP - ...

Syntax

TRAP ...

Format

The TRAP instruction uses the XXX instruction formats.

2	7	Reg R	0	M	1	Reg B	0	Reg X	0
2	4	4	1	1	1	4	2	4	9

Description

The TRAP instruction ...

The indexed formats may cause a memory reference type trap.

Operation

...

Exceptions

...

Notes

None.

XOR Boolean Operation

Syntax

```
XOR[.N] RegR, RegB, RegA
XOR[.N] RegR, RegB, SignedImmed15

OR[.N/D/W/H/B] RegR, SignedImmed13( RegB )
OR[.N/D/W/H/B] RegR, RegX ( RegB )
```

Format

The XOR instruction uses the register, the immediate, the indexed and the register indexed instruction formats.

0	4	Reg R	N	0	0	Reg B	0	Reg A	0
2	4	4	3			4	2	4	9

0	4	Reg R	N	0	1	Reg B	S-IMM-15
2	4	4	3			4	15

1	4	Reg R	N	0	0	Reg B	dw	S-IMM-13
2	4	4	3			4	2	13

1	4	Reg R	N	0	1	Reg B	dw	Reg X	0
2	4	4	3			4	2	4	9

Description

The XOR instruction performs the binary XOR operation on two 64-bit operands and stores the result to the target register **RegR**. The result can be negated by setting the "N" bit. The instruction supports the immediate, the register and the two memory reference instruction formats. The boolean operation itself does not cause any traps. The indexed formats may cause a memory reference associated trap.

Operation

```
RegR <- RegB ^ RegA (register format)
RegR <- RegB ^ immOperand( ) (immediate format)
RegR <- RegR ^ memOperand( ) (indexed formats)
if ( instr.[N] RegR <- not( RegR );
```

Exceptions

Data Alignment Trap
Memory Access Violation Trap
Memory Protection Violation Trap
Data TLB Miss Trap

Notes

None.

Appendix - Instruction Commentary

xxx

Appendix - Instruction Notation Routines

describe the routines used in the instruction set operation part...

Table 6: Instruction Notation Routines

Function	Description
...	...
...	...

Appendix -TLB and Caches

xxx

